



Society of Petroleum Engineers

SPE-226620-MS

Frequency Domain Characteristic of Acoustic Signals Generated by Proppant Slurry Flow Through Perforations

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Copyright 2025, Society of Petroleum Engineers DOI [10.2118/226620-MS](https://doi.org/10.2118/226620-MS)

This paper was prepared for presentation at the SPE International Hydraulic Fracturing Technology Conference and Exhibition held in Muscat, Sultanate of Oman, 23 - 25 September 2025.

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Abstract

This study focused on the frequency domain characteristics of acoustic signals generated from sand-perforation interaction. The effects of parameters on the spectrum of sand-related signals were discussed in detail. The goal is to predict the frequency range of sand-perforation interaction with given operation parameters. The predicted frequency range can serve as a reference when locating the sand-related signals from distributed acoustic sensors (DAS) data. With the frequency range determined, sand-related signals can be isolated at each cluster location and utilized to estimate sand distribution among clusters for each fracturing stage.

To determine the frequency domain characteristics of the sand-related signals, our approach combined the excitation frequencies and natural frequencies of a system to estimate the excited natural frequencies. We used the contact mechanics to estimate the contact time of sand and casing. With the contact time signal in the time domain, we used the Fourier transform to estimate the range of excitation frequency. The natural frequencies of a system can be determined from hammer tap tests and modal analysis with the finite element method. To estimate the natural frequencies of the system we used, both methods were conducted to the experimental apparatus. The estimated excited natural frequencies were compared to the sand-related frequencies determined in the slurry injection experiment to validate the approach.

The comparison between excited natural frequencies from the two methods and sand-generated frequencies from experiments shows that the spectra had a good agreement on the peak frequency location. Indicating that the approach developed can estimate the sand-related frequencies well. Based on the model of excitation frequency, sand sizes from 50 to 100 meshes have a minor effect on the sand-related frequency peak location but result in different excitation amplitudes. Parameters including impact velocity, sand density, sand radius, Young's modulus, and Poisson's ratio of both sand and casing determine the excitation frequency range. Casing material properties and corresponding boundary conditions (constraint and stress) determine the natural frequencies of the system. To apply this approach to the field DAS data interpretation, since hammer impact cannot be applied to the casing buried underground, a high-speed jet generated by a perforating gun was selected as the excitation source. By analyzing the DAS signal right after the perforating, when the casing still vibrates, the damped natural frequency can be captured from the frequency

domain. The captured damped natural frequency had an agreement with the sand-related signal determined during sand injection, which shows the possibility of estimating the range of the sand-related signal before sand injection.

This study provides a novel approach to predicting the sand-related frequency range resulting from sand-casing collision. This approach can be further applied to the field high-frequency DAS data to improve the interpretation result for sand distribution among clusters.

Introduction

During slurry injection, some clusters can receive more fluids than others and generate larger fractures, which results in other clusters receiving less slurry with insufficient fracture propagation. To maximize production, the industry tends to use completion designs with higher intensity over the years, and diversion technologies have also been applied to achieve uniformly distributed hydraulic fractures (Weijers et al. 2019), (Zhang et al. 2020). These trends in the industry are directed at maximizing the connection to low-permeability rocks. Ideally, contact points between the reservoir and well (fractures) are distributed everywhere along the wellbore.

The limited entry design is widely applied in completion design for hydraulic fracturing to achieve uniform fluid distribution among clusters, which results in a higher perforation pressure drop and requires a higher bottom hole treatment pressure. A common assumption in water/sand distribution estimation is that the sand distribution is the same as the fluid distribution. However, sand brings more uncertainty to the distribution estimation. Sand can cause significant erosion to the perforation hole. Erosion can serve as an issue in limited entry design by undermining its efforts to achieve uniform slurry distribution. Several field cases (Molenaar and Cox 2013), (Cramer et al. 2020) showed that the slurry is non-uniformly distributed even with a limited entry design. Sand distribution was also found to be different from fluid distribution.

With the different distributions between sand and fluids in limited entry design, a further estimation method is required for sand distribution. Distributed Acoustic Sensing (DAS) is introduced for hydraulic fracturing (Ugueto et al. 2015, Ugueto et al. 2016) and used for sand distribution estimation (Wheaton et al. 2016). In recent years, DAS has been widely used in hydraulic fracturing monitoring for water/sand distribution in in-well monitoring. Shen et al. (2017) coupled perforation erosion with DAS measurement to better understand water and sand distribution in different operation periods. Zakhour et al. (2021) selected certain frequency bands for estimating water and sand distribution. To further understand the features of the acoustic signal when slurry flows through perforation and explore the contribution of water and sand to the overall acoustic signal, Chen (2025a) conducted sand transportation experiments through perforation under a lab-scale limited-entry design. By comparing the experimental data recorded during the only water pumping period and injection period of 0.1 ppg of slurry, sand-related signals were distinguished from water flow sounds in the frequency domain. Certain frequency bands picked up when sand is introduced to the system. Two frequency ranges enhanced by sand were 1000 to 2000 Hz and 3000 to 5500 Hz. With further comparison between different sand concentrations from 0.15 to 0.19 ppg, the magnitude of frequencies between 2700 to 4000 Hz were determined to be proportional to sand concentration. With regression analysis, sand concentration and averaged signal amplitude were determined to be linearly related.

To further understand whether sand concentration changes the amplitude of the related frequency peak or results in a related frequency shift, simulations were conducted based on the vibration theory. Vibration of the casing system after the impact of a sand particle was modeled by an underdamped vibration model. The acoustic signal generated from multiple sand particles impacting the casing was analog by the superposition of the sound signal generated from individual vibrations. The result of the superposition shows that the number of superposition (sand concentration) has no effect on frequency peak locations; frequency peak amplitude increases with increasing sand concentration, which agrees with the experimental observation on

the frequency range from 2700 to 4000 Hz. Experimental results show that introducing sand enhances the amplitude of certain frequencies while leaving the rest of the frequencies relatively unchanged.

Chen et al. (2025b) applied the observation from the experimental study to field data. Signal processing methods were applied to the high-frequency DAS data for multiple clusters in the time domain, spectrograms (frequency domain waterfall plots) were obtained at each cluster location. Major frequency components generated by sand injection were identified for major clusters from the comparison between the water injection period and two slurry injection periods. Two methods were applied to estimate the sand distribution among the clusters using the amplitude of sand-generated frequency bands.

While Chen (2025a) focused on experimental study and Chen et al., (2025b) applied the lab observation to field data, this study developed an approach to understand the acoustic wave generated during the impact of sand and casing. With the established method using sand-related frequency peak amplitude to estimate sand distribution, this study provided an approach to estimate the location of sand-related frequency peaks.

Signal processing methods (Oppenheim and Schaffer 1977) and theories of elasticity (Timoshenko et al., 1970) were used to develop the approach. Results obtained from the experimental study (Chen 2025a) were used to validate the approach. Results from Wang et al., (2019) were used to verify the excited frequency model in our approach. Combining the concepts of excitation frequencies and natural frequencies of a system, the acoustic signal characteristics for different sand diameters, or sand/casing combinations with various material properties were discussed in detail.

Determine Vibration Frequency from Excitation and Natural Frequencies

With the frequency peak locations not affected by sand concentration (Chen 2025a), one question that remains is what factors determine the frequency components for a sand-metal impact signal.

When slurry flows through perforations, sand particles continuously impact the casing around perforations, and the collision results in vibration. The vibration generates a sound/pressure wave. The sound wave is then captured as part of the acoustic signal. For the sand-casing collision case, the excitations are a series of discrete events. After each excitation, the system has free vibration at its natural frequency. To understand what vibration and sound can be triggered from the casing system, it is necessary to determine what excitation can result from impact with sand particles and what the natural frequencies of a casing system are.

Obtaining the contact time of sand-casing collision is the first step to estimating the excitation frequency from a collision. Hertz (1896) provided a theoretical model to estimate the contact duration during the collision between two spheres, which can be further modified to understand the particle and plate collision, a detailed derivation was included in Appendix A. Hertz contact theory assumed that contact process is purely elastic under a quasi-static condition. During the collision, energy dissipation was ignored and elastic mechanics equations can be directly applied.

With two assumptions, the contact time and the peak can be calculated based on the material properties and geometries of the casing and the sand particle. The contact duration is expressed as (Hertz 1896):

$$t_{el} = 2.94 \left(\frac{5\pi}{4} \right)^{\frac{2}{5}} \rho^{\frac{2}{5}} \frac{r}{V^{\frac{1}{5}}} (f(E))^{\frac{2}{5}} \quad (1)$$

$$f(E) = \frac{1-\nu_1^2}{E_1} + \frac{1-\nu_2^2}{E_2}$$

Where t_{el} is the elastic impact time in microseconds, ρ is the sand density in g/cm³. r is the particle radius in μ m. V is the impact velocity in m/s. $f(E)$ is the effective contact modulus in kPa. ν_1 is the Poisson's ratio for the casing and ν_2 is the Poisson's ratio for the sand particles. E_1 is Young's modulus for the casing in kPa and E_2 is Young's modulus for sand particles.

The impact time function is the key to determining the range of excitation frequencies. From the Fourier transform, a short impact in the time domain results in an excitation with a wide frequency range. For the elastic collision, the contact process is assumed to be symmetric with a maximum contact force F_{el} . By directly connecting three points which represent contact start, maximum contact and contact end, an isosceles triangular impulse can be obtained to describe the time and contact force relationship during a collision, and the excitation spectrum can be estimated from a given isosceles triangular impulse. The left side of Figure 1 shows an example of a triangular impulse with a duration of t_{el} .

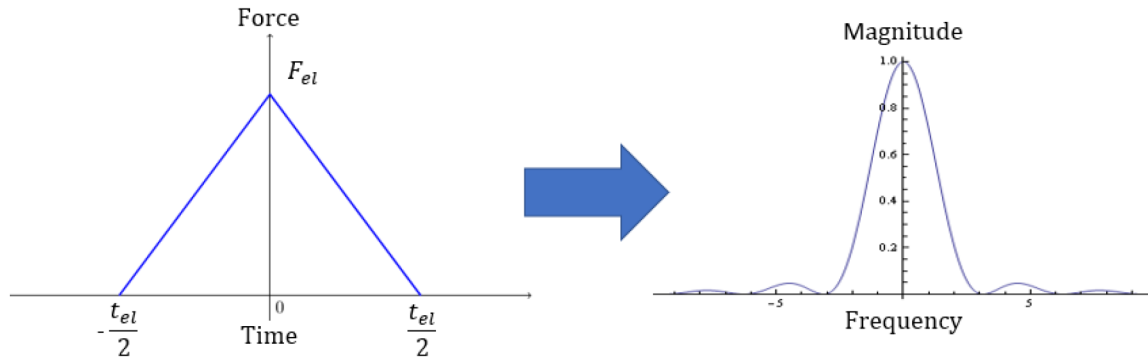


Figure 1—Impact in the Time Domain and Frequency Domain

At the first point when $t = -\frac{t_{el}}{2}$, just before the sand particle-casing impact, the sand particle has the highest velocity towards the casing. At the second point when $t = 0$, the velocity of the sand particle decreases from the initial impact velocity to zero, all the kinetic energy is transferred as the strain energy, and the contact force (stress) reaches the maximum as the deformations (strain) reach the maximum level. After the second point, sand gets bounced back, and the strain energy transfers back to the sand particle, causing a velocity in the opposite direction as the initial impact, causing the sand particle to move away from the casing. At the third moment when $t = \frac{t_{el}}{2}$, the particle has no contact with the casing and the contact force decreases to zero. With the time domain signal as a triangular impulse, the spectrum is a sinc^2 function, which is expressed as:

$$F(\omega) = \frac{A' \tau'}{2} \left(\frac{\sin \frac{\omega \tau'}{4}}{\frac{\omega \tau'}{4}} \right)^2 = \frac{A' \tau'}{2} \text{sinc}^2 \frac{\omega \tau'}{4} \quad (2)$$

Where A' is the peak amplitude of the triangular impulse, τ' is the duration of the triangular impulse. For a collision, τ' is the total contact time t_{el} in Equation 1. With given amplitude and duration, using the property of the sinc^2 function, the major range from excitation can be solved as $\frac{A' \tau'}{2} \text{sinc}^2 \frac{\omega \tau'}{4} = 0$, which is the point when the excitation amplitude $F(\omega)$ becomes zero. Defined that the frequency limit of the major lobe is obtained when $F(\omega)$ first becomes zero. Based on equation 2, solving $\frac{A' \tau'}{2} \text{sinc}^2 \frac{\omega \tau'}{4} = 0$ can be simplified by solving $\sin \frac{\omega \tau'}{4} = 0$, let $\frac{\omega \tau'}{4} = k\pi$, with $k=1$ for the major lobe, the maximum angular frequency can be expressed as $\omega = \frac{4\pi}{\tau'}$, with the angular frequency $\omega = 2\pi f$, the main lobe excitation frequency is $f = \frac{2}{\tau'}$, which indicates that a shorter impact has a higher excitation frequency. With this equation, the main lobe excitation frequency can be calculated easily with a given impact time τ' . By combining the Hertz contact time equation with the first zero-point calculation, the effect of casing and particle properties can be observed directly. Recall the contact time from elastic collision:

$$\tau(t_{el}) = 2.94 \left(\frac{5\pi}{4} \right)^{\frac{2}{3}} \rho^{\frac{2}{3}} \frac{r}{V^{\frac{1}{3}}} \left(\frac{1-v_1^2}{E_1} + \frac{1-v_2^2}{E_2} \right)^{\frac{2}{3}} \quad (3)$$

With the relationship solved from excitation frequency $f = \frac{2}{\tau}$:

$$f = \frac{2}{\tau} = \frac{2}{t_{el}} \quad (4)$$

By combining [Equations 4 and 6](#), the excitation frequency from a triangular impulse can be expressed as:

$$f = \frac{V^{\frac{1}{3}}}{1.47 \left(\frac{5\pi}{4} \right)^{\frac{2}{3}} \rho^{\frac{2}{3}} r \left(\frac{1-v_1^2}{E_1} + \frac{1-v_2^2}{E_2} \right)^{\frac{2}{3}}} \quad (5)$$

Instead of simplifying the force impulse with a triangular impulse, [McLaskey and Glaser \(2010\)](#) summarized two force impulse models and their frequency responses. The first force impulse model was developed by [Hunter \(1957\)](#), which is a concave portion of a sine function in one period (half-sine). The second force impulse model was developed by [Reed \(1985\)](#) by modifying Hunter's correlation, which used $(\sin x)^{1.5}$ to replace the original sine function. Same as the triangular impulse, two force impulse models are symmetric in the time domain. To understand the effect of different force impulse models on excitation frequency range estimation, Fourier transform of the half-sine force impulse was calculated for comparison. The half-sine impulse was regarded as the cosine impulse with a phase shift. The target impulse with duration of τ can be obtained from a continuous cosine function with a period of 2τ , such a cosine function has an angular frequency is $\omega = 2\pi/2\tau = \pi/\tau$, with a given amplitude of A , the cosine signal has a function of:

$$f(t) = A \cos \frac{\pi}{\tau} t \quad (6)$$

The Fourier transform half-sine impulse can be further expressed as:

$$F(\omega) = A \int_{-\tau/2}^{\tau/2} \cos \frac{\pi}{\tau} t e^{-j\omega t} dt = A \left[\frac{\sin\left(\frac{\omega\tau - \pi}{2}\right)}{\omega - \frac{\pi}{\tau}} + \frac{\sin\left(\frac{\omega\tau + \pi}{2}\right)}{\omega + \frac{\pi}{\tau}} \right] \quad (7)$$

To solve the main lobe excitation frequency limit, with the same procedure, the goal is to calculate the first point when $F(\omega)$ become 0. As discussed, a sine function can be regarded as cosine function with a phase shift, therefore $\sin\left(\frac{\omega\tau - \pi}{2}\right) = -\cos\frac{\omega\tau}{2}$ and $\sin\left(\frac{\omega\tau + \pi}{2}\right) = \cos\frac{\omega\tau}{2}$. With $\cos\left(\frac{\pi}{2} + k\pi\right) = 0$, let $\frac{\omega\tau}{2} = \frac{\pi}{2} + k\pi$, the main lobe corresponds to $k=0$, therefore $\omega\tau = \pi$ and the main lobe excitation frequency limit was solved as and

$$\omega = \frac{\pi}{\tau} \quad (8)$$

$$f = \frac{1}{2\tau} = \frac{1}{2t_{el}}$$

By combining [Equations 1 and 8](#), excitation frequency can be expressed as:

$$f = \frac{V^{\frac{1}{3}}}{5.88 \left(\frac{5\pi}{4} \right)^{\frac{2}{3}} \rho^{\frac{2}{3}} r \left(\frac{1-v_1^2}{E_1} + \frac{1-v_2^2}{E_2} \right)^{\frac{2}{3}}} \quad (9)$$

Compared to the triangular impulse, the half-sine impulse has a shorter excitation frequency range. The difference between two waveforms is $\Delta f = \frac{3}{2} t_{el}$. By plotting the spectra generated by [Equations 2 and 7](#), the spectrum of a half-sine force impulse has a flatter shape than the triangle force impulse, in which the excitation amplitude decreases at a slower rate with higher frequencies on the spectrum. Based on

comparison, different assumptions on the force impulse resulted in different shapes of the spectra and the excitation frequency limitations (locations for the first zero point).

From both [Equations 5 and 9](#), it can be observed that excitation frequency increases with higher collision velocity, decreases with larger density and radius of the sand particles, and increases with larger Poisson's ratios and modulus of both sand and casing. Since elastic collision results in a short contact time and a wide excitation frequency, it can also be speculated that the contact time or excitation frequency is determined by the efficiency of the transfer of kinetic energy during collision (including the amount of deformation and energy dissipation), a higher efficiency resulting in a higher excitation frequency. Liquid and solid have a large difference in the kinetic energy transfer efficiency during collision, as discussed previously, the ideal sand particle-casing collision case was assumed to be purely elastic without energy dissipation, sand particles have a negligible amount of deformation; in contrast, water droplets cannot maintain their shape during collision and have more energy dissipation. During contact, large deformation of the droplet increases the contact area, decreases the average contact force and results in a longer contact time during water droplet-casing collision. This is the essential reason that the signals generated by a water droplet impacting the casing and a sand particle impacting the casing are different.

With a known excitation frequency range, the next step is to estimate the natural frequencies of the system. Natural frequency is a general property of a system since a system has free vibrations at its natural frequencies with a transient excitation/ initial disturbance. For a simple one degree of freedom (DOF) mass-damper-spring system with mass m , stiffness k , and viscous damping constant c , with an excitation amplitude of F , the static deflection is defined as ([Ingard 1988](#)):

$$\delta_{st} = \frac{F}{k} \quad (10)$$

The ratio between the output complex amplitude $|X|$ from the system with the input δ_{st} can be expressed as:

$$\frac{|X|}{\delta_{st}} = \frac{1}{\sqrt{(1-r^2)^2 + (2\zeta r)^2}} \quad (11)$$

Where frequency ratio $r = \omega / \sqrt{\frac{k}{m}}$ and damping ratio $\zeta = \frac{c}{\sqrt{4mk}}$. [Equation 11](#) is known as the Frequency

Response Function (FRF) for a system with vibration. The hammer tap test is a common method to determine the frequency response function for a physical system in the lab.

The hammer tap test uses a hammer to hit the system to create a short impact as excitation. As discussed, a short time excitation in the time domain is equivalent to a wide frequency excitation in the frequency domain. All the natural frequencies of the system within the excitation range will be triggered. From the frequency response function, when the excitation frequency matches the natural frequency, frequency ratio r becomes 1 and the frequency response function becomes $\frac{1}{2\zeta}$, which gives a local maximum value. By recording the response of the system after the excitation, we can estimate the natural frequencies of a system by identifying the peak frequencies in the response spectrum function.

In engineering applications, it is rare for a system to have only one DOF. For a mass-damper-spring system with multiple DOFs, mass, damping, and stiffness are expressed as matrices. Matrices are assembled by the connection between individual DOF in the system. Mode shape is the shape of a structure when it vibrates at a specific natural frequency. In a multiple DOFs system, each natural frequency corresponds to one mode shape and one DOF. Another common approach to estimating the natural frequencies and corresponding mode shapes is the finite element method. A continuous system has an infinite number of natural frequencies and mode shapes. However, for most of the structure, lower natural frequencies (for example, natural frequencies related to the first five mode shapes) are more important since they are related to the basic mode shape which can be excited easily. Therefore, a finite element model with a finite

number of natural frequencies can be used to estimate the first few natural frequencies. Elements used in the finite element model are usually 3D elements. The total DOFs of an element can be determined from the production of the node number and DOF per node.

Case Study: Acoustic Behavior of Particles through the Orifice

With the theoretical approach of combining excitation frequencies with the natural frequencies of a system, data from the experiments can be used to validate this approach for a more complex condition.

Determine the Excitation Frequency

Based on Equation 1, material properties and the size of the particle are required to estimate the contact time for collision between casing and sand particle under the lab conditions.

In the experiment (Chen 2025a), collision mainly happened on the disk, which is made of 17-4 stainless steel. The steel has a Young's modulus of 28500 ksi, which is equivalent to 196.5 GPa. The Poisson's ratio is around 0.272. Density is around 0.282 lb/in³, equivalent to 7.81 g/cm³. Young's modulus of sand can vary based on the sand types; the sand used is assumed to have the same Young's modulus as fused quartz of 71.7 GPa and a Poisson's ratio of 0.17. The density of the sand is assumed to be 2.65g/ cm³. The top three sizes from the sieve test were 50 mesh, 70 mesh, and 100 mesh, with mass percentages of 34%. The impact velocity used can be assumed to be the flowing velocity before the collision, a typical velocity from the experimental study is 0.66m/s.

By using Equation 1, contact times for 50 mesh (297 μ m ID), 70 mesh (210 μ m ID), and 100 mesh (149 μ m ID) sands are estimated to be 0.97, 0.69, and 0.49 μ s. Assuming the contact process is described by a triangular impulse, by using Equation 5, the corresponding first zero points of the excitation frequency are 2058, 2910 and 4102 kHz. Assuming the contact process is described by a half-sine impulse, by using Equation 9, the corresponding first zero points of the excitation frequency are 514, 728 and 1025 kHz. As discussed previously, assumptions about the force impulse can result in different excitation frequency ranges. Despite the difference between the excitation frequency ranges estimated by two impulse models are apparent, the measured excited frequencies can be very close. The Nyquist frequency limit of lab measurement was 25.6 kHz and the interested frequency range was 0 to 8 kHz (Chen 2025a). The excitation frequency range estimated by either impulse model is far beyond the Nyquist frequency limit and the interested frequency range. With large excitation frequency ranges, the frequency range difference between impulse models and the difference between particle sizes become less significant, therefore all the measurable natural frequencies of the system can be excited by the sand-metal collision with sands in three different sizes. In other words, to current system and measurement equipment, selections on the impulse models or particle sizes difference (50, 70, and 100 meshes) have equivalent excitation frequency in the interested frequency range.

Research done by Wang et al., (2019) showed a consistent result with this conclusion. One of their lab experiments recorded the sound and vibration signals generated by the collision between the sand particles and a metal plate. In the experiment, sand particles are released at the same height above the plate, using gravity to generate the same impact velocity that is perpendicular to the plate. Wang et al., (2019) used sand particles of five different sizes in their experiment. The particles have a size range from 150 to 300 μ m, which is equivalent to 50 to 100 meshes. Figure 2 shows the comparison of the recorded spectra generated by sand with different sizes.

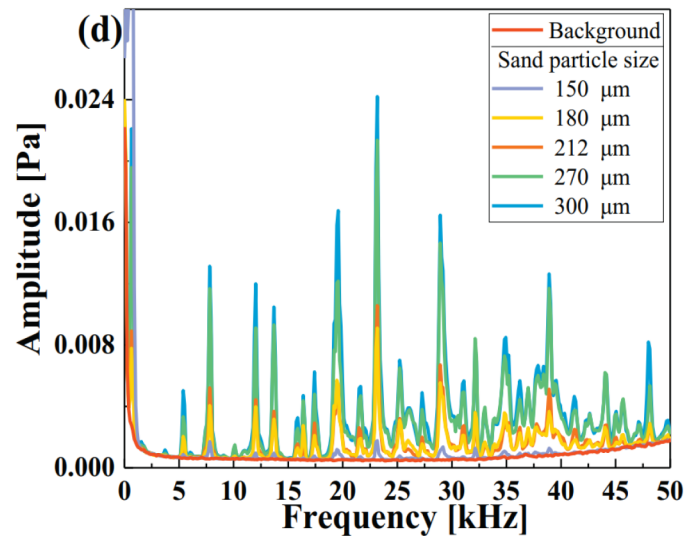


Figure 2—Spectrum Functions Generated by Sand Impact with Different Sizes (Reprinted from Wang et al., 2019 Figure 4d)

The result shows that spectra generated by sand particles with five different sizes are similar in shape within the frequency range from 0 to 50 kHz, which agrees with the previous conclusion that the excitation frequency range is far beyond the measured frequency range, and the difference between models and particle sizes (50 to 100 meshes) becomes insignificant. Another observation from Figure 2 is that the larger particles generated a frequency peak with a larger amplitude. With the same velocity from free fall, heavier particles have more kinetic energy and generate stronger acoustic waves, which indicates a higher energy transformation level between a sand particle and a plate resulting in higher amplitude in both excitation and response.

Natural Frequency: Lab Hammer Test

With a known excitation range, both the hammer tap test and frequency analysis using the finite element method are applied to estimate the Frequency Response Function. For the experiment apparatus in the lab, the casing-perforation system consists of one cell as the casing and a circular metal plate with an orifice (disk) inserted inside the cell as the perforation. Other parts inside the test cell include the extended pipe, the polymer spacer in front of the disk and the core spacer behind the disk. All these parts contribute to the distribution of mass and stiffness inside the test cell. In an ideal hammer test, impact should be applied to the disk with all parts assembled. However, the caps of the test cell need to be removed to apply direct impacts to the disk. For a simplified lab hammer test, the test cell is disconnected from the entire circulation system after a set of slurry injections. Impacts from a steel hammer are applied only to the metal plate and the cell body under assembly. The microphone was set to be perpendicular to the test cell body, which has the same setup as the slurry sound measurement discussed in Chen (2025a).

During slurry injections, the flow trajectories of sand particles converge near the orifice, particles then flow through the orifice while having collision with the metal disk. During collision, sand particles applied impacts to the disk in the perpendicular direction in a circular region around the orifice. To replicate the sand particles' impact process, perpendicular impacts were applied to multiple points in the region near the orifice with a steel hammer during the tap test. Before the tap test started, sand particles from previous slurry injections were embedded into the micro gap between the disk and the inner surface of the test cell, which further held the disk in place during the hammer impact. Besides the impacts on the disk, impacts were also applied to the test cell body (cylinder part) to understand the total response for an assembled cell-orifice system.

For signal analysis, a complete signal from the starting point of each signal results in waveform decay to zero or the background noise level was used. The hammer tap test for the disk included eight impact

events. The time domain signal for each impact was carefully extracted from the original record, and the spectrum of each impact was compared. Figure 3 shows the time domain acoustic signal during the impact test. Figure 4 shows a zoomed-in view of impact No.4 as an example.

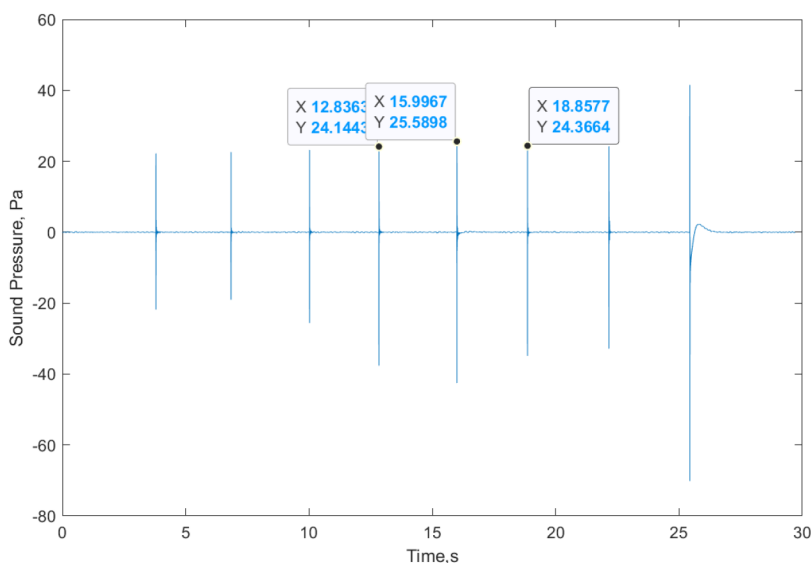


Figure 3—Time Domain Acoustic Signal for Impact on Disk

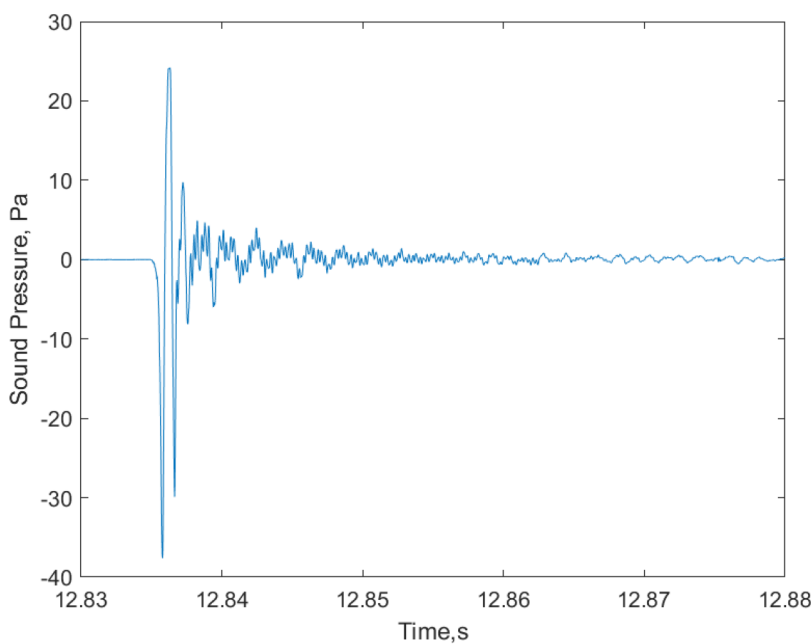


Figure 4—Detail of Impact No.4

It can be observed that hammer impact introduces a peak in the sound pressure measurement, which was summarized as the transient excitation/ initial disturbance in the previous discussion. After the impact, the response signal oscillates and decays with time, which is an underdamped condition as discussed in Chen (2025a). Figure 5 shows the spectrum functions of response signals for 4th(left), 5th (middle) and 6th (right) impacts individually.

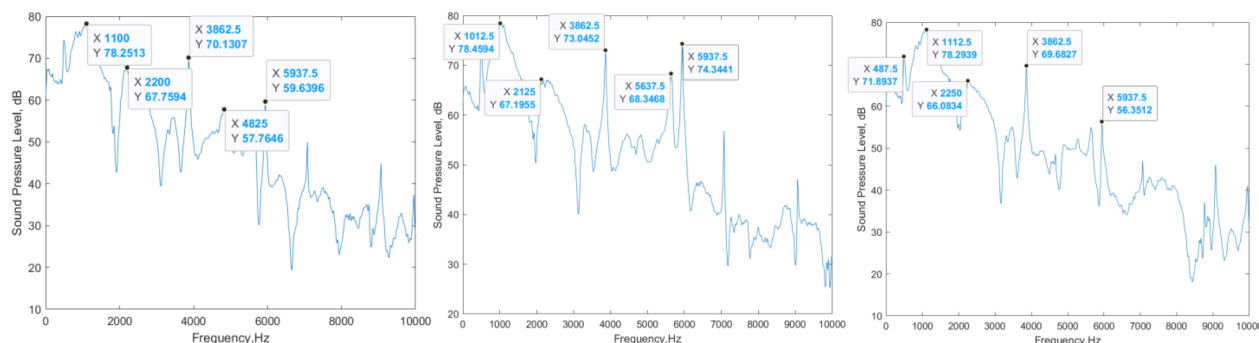


Figure 5—Frequency Response Function Generated from Impact No.4, 5 and 6 on Disk

Several peaks can be identified in Figure 5, which represent the damped natural frequencies excited by the hammer impact. It can be observed that the spectrum functions from the three spectra share most of the peak locations, which indicates the measurements are consistent. Figure 6 shows the comparison of three spectra from impacts on the disk.

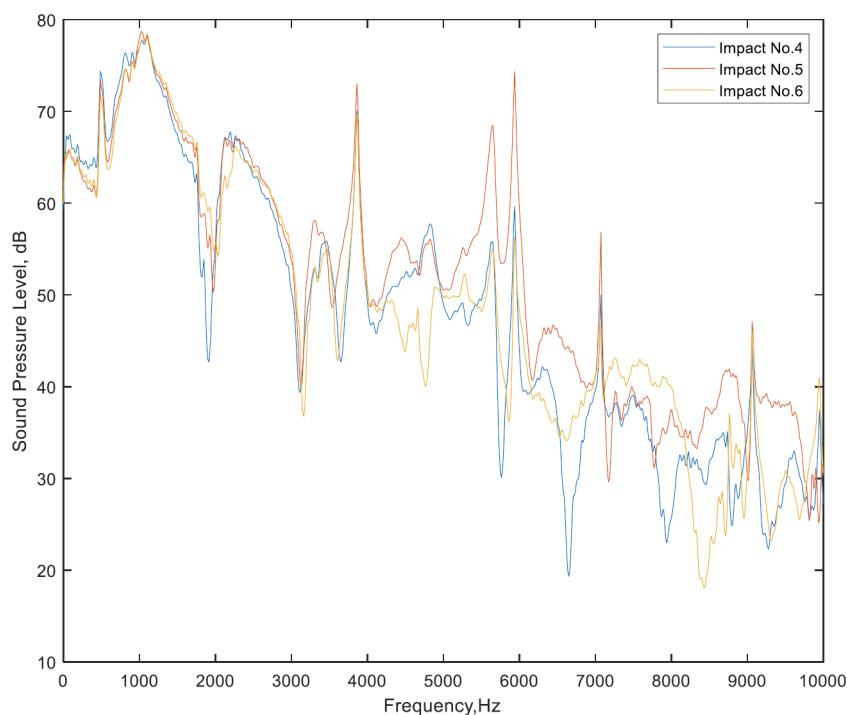


Figure 6—Comparison of Spectra Generated from Impacts on Disk

From Figure 6, locations of some frequency peaks still have minor differences, whether a mode can be triggered also depends on the position of the excitation. Although all impacts happened in a small zone near the orifice, each impact happened in a different position. Excitations can be applied to the node of a certain mode shape in some impacts, which results in a minor difference between the spectra. From Figures 5 and 6, common frequency peaks can be identified at around 3862 and 5937 Hz, these peaks have sharp shapes in the spectrum. Two wider peaks can be found at 1100 and 2200 Hz, which are much flatter compared to other peaks.

As mentioned, the hammer tap test was also conducted on the inner surface of the cell with the cell-orifice system assembled. Figure 7 shows the time domain signal of impacts. Five impacts were applied to the inner surface.

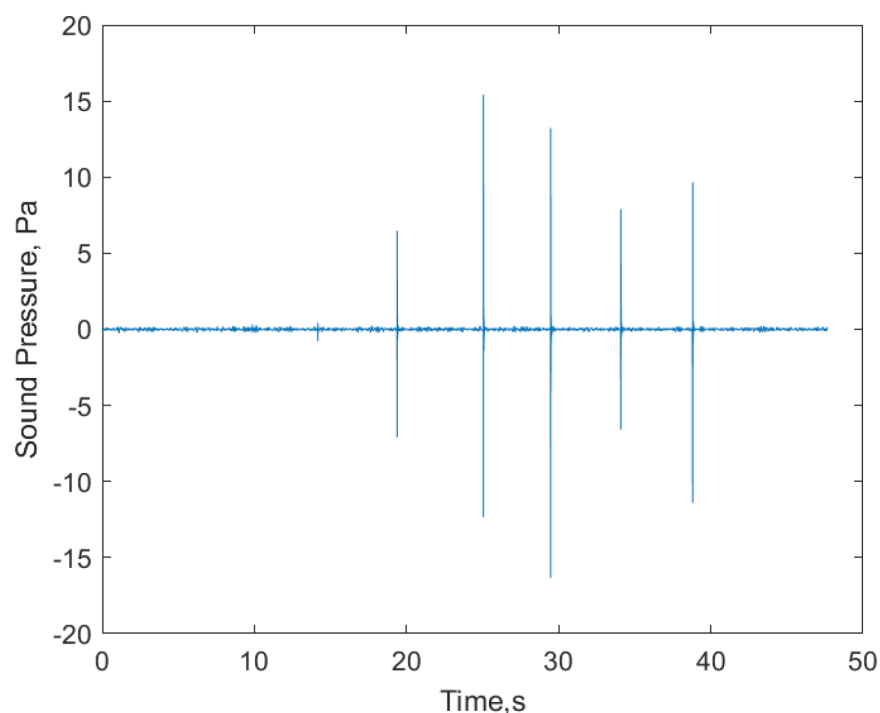


Figure 7—Time Domain Acoustic Signal for Impact on Cylinder

Figure 8 shows the response spectrum functions for 2nd (left), 4th (middle), and 5th (right) impacts with peak locations marked. Figure 9 shows the comparison of three spectra from impacts to the cylinder. The impact positions are widely distributed on the outer surface of the cylinder, resulting in less consistent frequency response functions for the cylinder than the disk. Despite the shapes of the spectrum functions that vary with impact, common frequency peaks were obvious, which appear in 1712, 2125, 3887, and 7137 Hz.

By comparing the common frequency peaks for the disk and test cell body (cylinder), some frequencies observed in the disk impact test also appear in the cylinder impact test with a slight frequency shift. For example, the frequency peaks of 3862.5 Hz shown in the disk and 3887.5 Hz from the cylinder are very close, showing excitation at different locations resulting in different responses for the assembled system used in the experiment. For the field case, the slurry flows through the perforation and excites the surrounding area. Different from the lab apparatus that consists of two parts, vibration initiates on the casing near perforations. Frequency response functions should be more consistent when excitation is applied to locations within a small area.

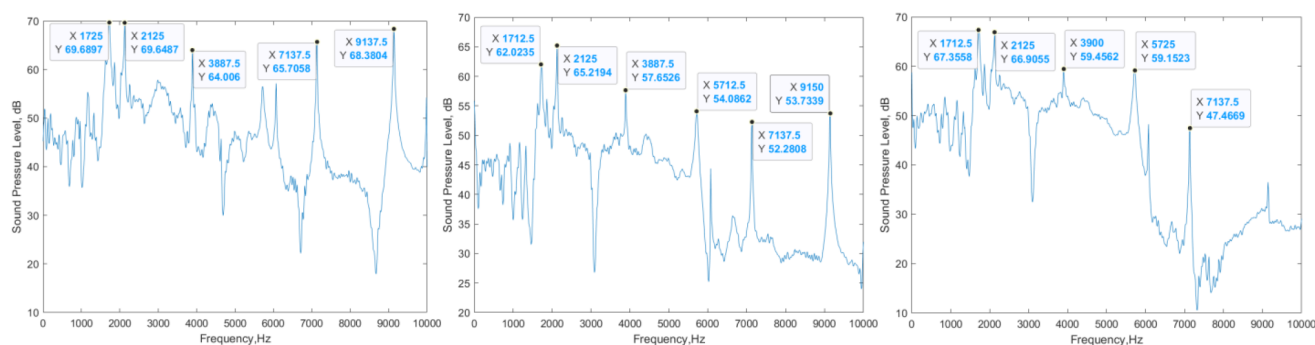


Figure 8—Cylinder Frequency Response Function Generated from Impact No.2, 4 and 5 on Cylinder

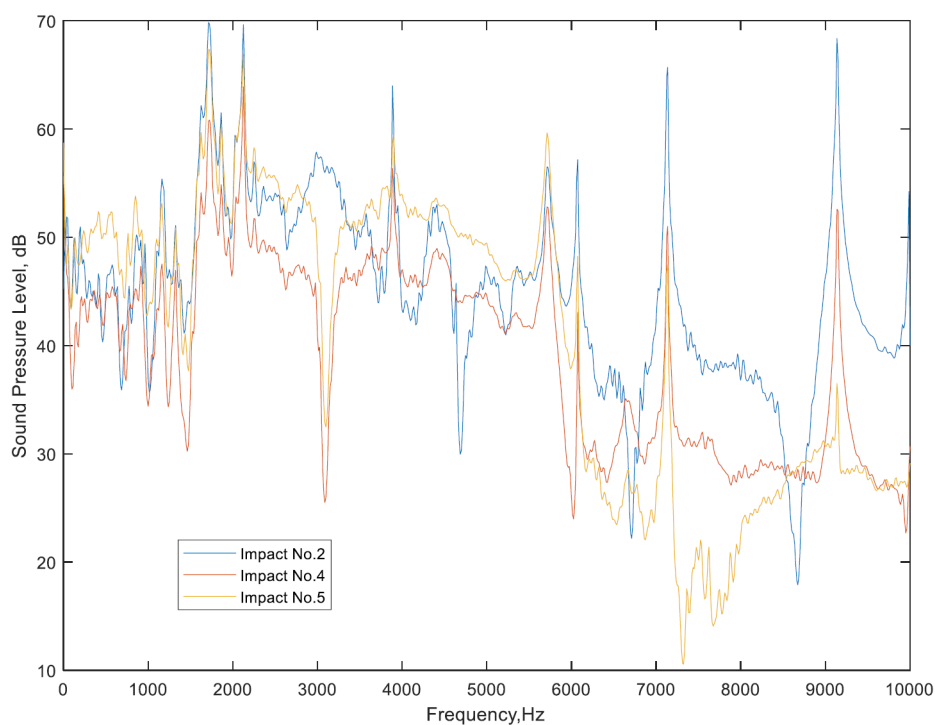


Figure 9—Comparison of Spectra Generated from Impacts on Cylinder

Natural Frequency: Finite Element Method

For frequency analysis using the finite element method, a disk is created in CAD software with the same geometry and material as used in experiments (Chen 2025a). Table 1 shows the material properties of the disk.

Table 1—Material Properties Used in Finite Element Model

Property	Value	Units
Elastic Modulus	28500000	psi
Poisson's Ratio	0.272	N/A
Shear Modulus		psi
Mass Density	0.282	lb/in ³
Tensile Strength	185000	psi
Compressive Strength		psi
Yield Strength	160000	psi
Thermal Expansion Coefficient		/°F
Thermal Conductivity	10.6	Btu/(in·sec·°F)
Specific Heat		Btu/(lb·°F)
Material Damping Ratio		N/A

In the experiment, the disk was inserted into the cylinder, as shown in Figure 10.



Figure 10—Disk inside the Cylinder

Since the disk and cylinder are not in a rigid connection, the disk can still move freely inside the cylinder along the axial direction. For the experimental setup, the core spacer holds the disk in place from the other side. For a simplified finite element model, only the disk was built and simulated. In a cylindrical coordinate system, the disk is allowed to have axial translation and circumferential rotation. The only constraint applied is that the disk cannot expand in a radial direction since it is restricted by the inner surface of the cylinder. Figure 11 shows the constraints applied to the disk in the natural frequency simulation.

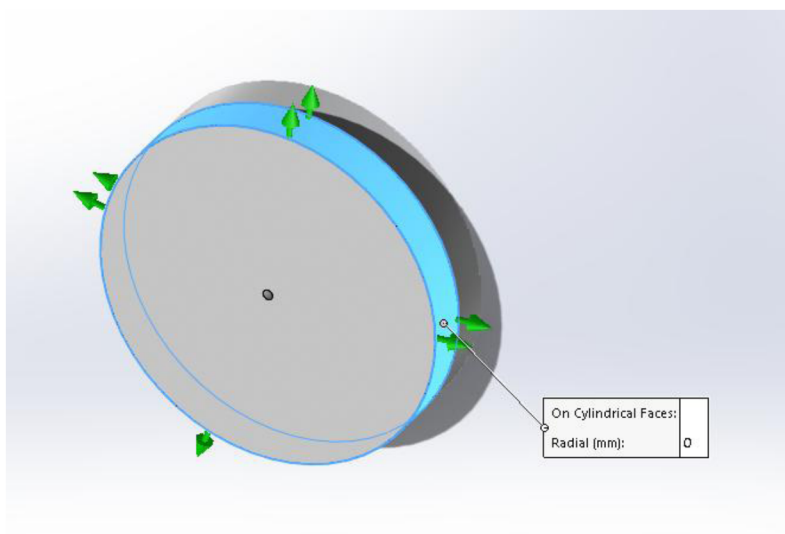


Figure 11—Boundary Conditions Applied on the Disk: Restricted Radial Expansion

No external load or excitation is applied for natural frequency analysis. The element size used for the 3.85 inches ID disk is 0.094 in. The element size is confirmed by checking the mesh convergence. Figure 12 shows the mesh detail.

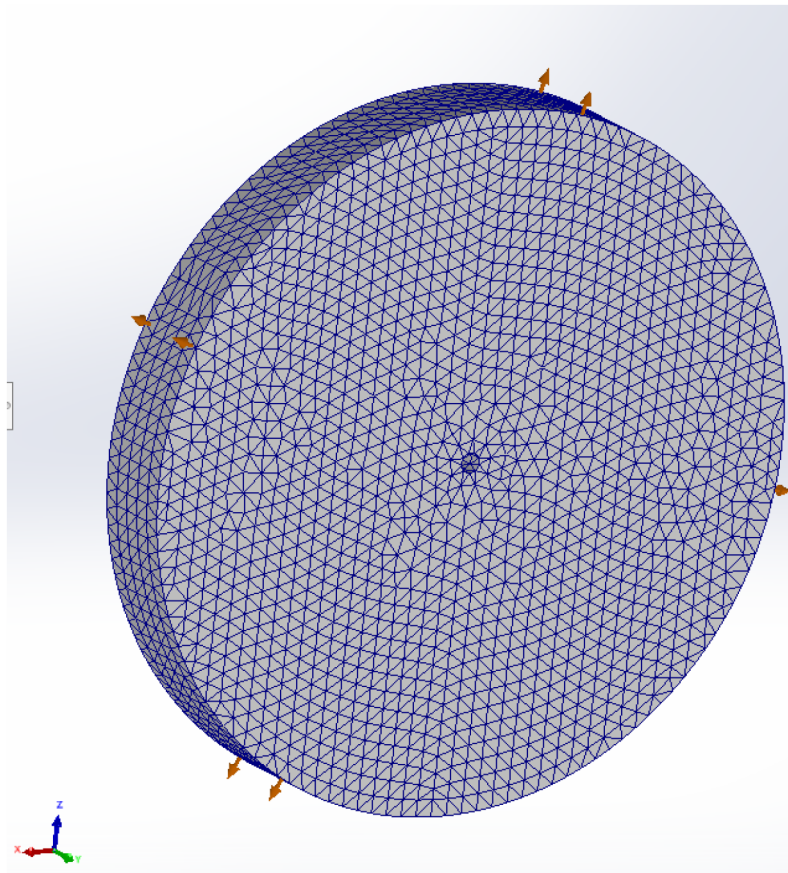


Figure 12—Mesh Created with 0.094 in. Elements

Table 2 shows the list of natural frequencies calculated from the finite element simulation.

Table 2—Model Mode Shapes and Corresponding Angular Frequency, Frequency, and Period

Mode No.	Frequency(Rad/sec)	Frequency(Hertz)	Period(Seconds)
1	0	0	1e+32
2	0.0087536	0.0013932	717.78
3	23,485	3,737.7	0.00026755
4	23,492	3,738.9	0.00026746
5	64,045	10,193	9.8106e-05

Since there is no constraint on the disk for axial movement, the first mode shape is the rigid body motion. The disk has no deformation in this mode shape. The corresponding natural frequency is zero. The second mode shape represents the shrinkage and expansion of the disk inside the cylinder. This process takes around 718 seconds, which is a long time, and the corresponding natural frequency is small. Third and fourth mode shapes have similar vibration patterns but in different directions. Corresponding natural frequencies are 3737.7 and 3738.9 Hz for the third and fourth mode shapes. Figure 13 shows the corresponding mode shapes to natural frequencies 3737.7 Hz (left) and 3738.9 Hz (right).

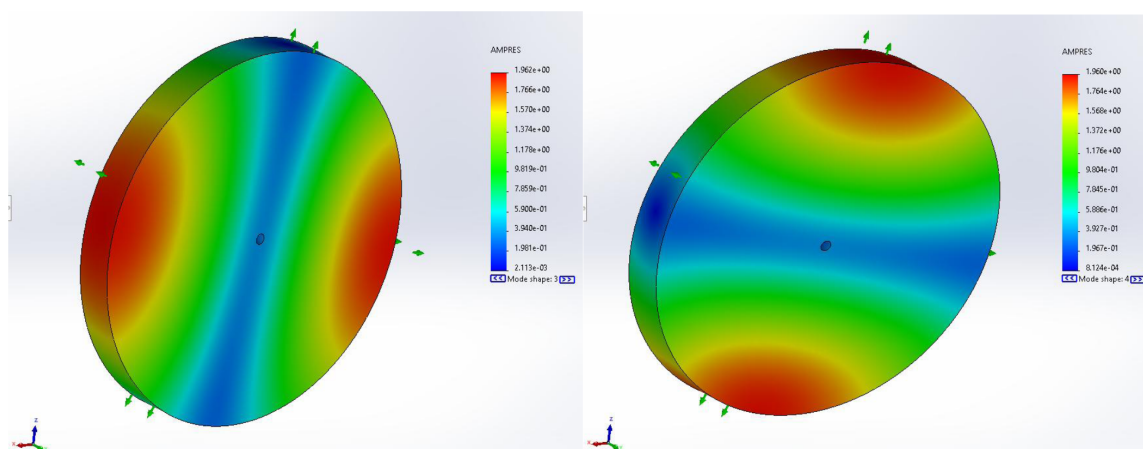


Figure 13—Third (Left) and Fourth (Right) Mode Shapes of the Disk

By comparing the natural frequencies from the tap test and finite element simulation, the simulation results do not show as many frequency peaks as the hammer impact test. Possible reasons include:

1. The tap test is conducted on the assembled system. However, only the disk is modeled and simulated in the finite element analysis. An assembled system has more DOFs than a disk. These DOFs and corresponding natural frequencies are invisible in the simulation result.
2. Only one constraint is applied in the simulation. Boundary conditions in reality are more complex with coupling with the core spacer and other devices inside the test cell, which change the distribution of stiffness and mass and result in differences in the disk's natural frequencies.

Result Comparisons: Hammer Test, Finite Element Method and Slurry Spectra

Despite the fact that the tap test shows a more complex feature than the simulation result, both methods have a general agreement on the location of a major frequency peak around 3800 Hz. The tap test for the disk shows a frequency peak of 3862 Hz and the simulation has a frequency peak of 3738 Hz. With agreement between the hammer test and the finite element simulation, the spectrum functions from the slurry flow experiment can be added to the comparison. Figure 14 focuses on the sand-related peak locations. The legend on the upper right side represents the sand concentration of slurry in pound per gallon (ppg).

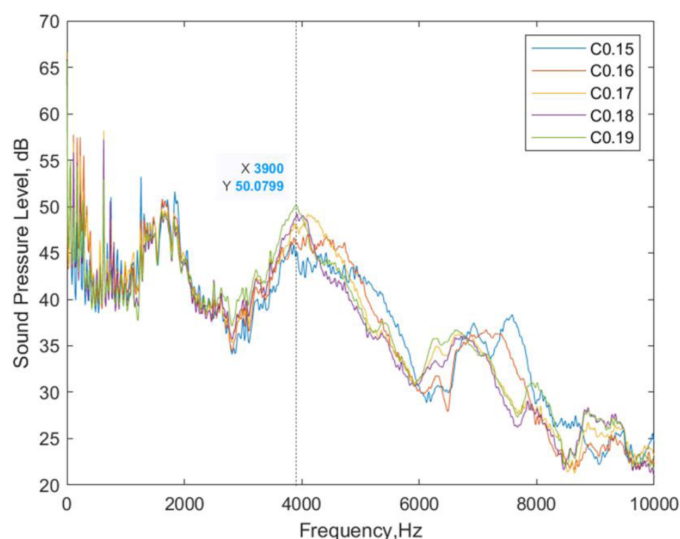


Figure 14—Frequency Peaks Location and Frequency

From Figure 14, the dashed line marks the peak location of the 0.19 ppg slurry's spectrum, spectra for all five concentrations have a major peak frequency of between 3800 and 3900 Hz. From the comparison, a common frequency peak around 3800 Hz was found in both slurry signal spectrums and natural frequency analyses. This finding indicates that the collision between sand and the metal disk-cylinder system excited the natural frequencies of the system. However, the spectrum function from the slurry flow experiment contains more frequency components, indicating several different sources contributing to the overall acoustic signals. From Figure 14, pump induced acoustic signal has a narrow spikes below 1200 Hz. Turbulent flow also contributes as a potential sound source, which generates eddies and acoustic signals.

With frequency content confirmed at around 3800 Hz, the vibration waveform for single particle impact can be expressed as:

$$x(t) = Ae^{-\beta(t-\Delta t)} \sin[3800 \cdot 2\pi(t - \Delta t)] \quad (12)$$

A is the amplitude corresponding to signal particle impact, t is the time, Δt is the time interval between two impacts.

For a multiple particles impact case, as discussed in Chen (2025a), the summation of impacts can be expressed as:

$$x_{total}(t) = \sum_{i=1}^{N_{sand}} x(t) \quad (13)$$

N_{sand} is the total particle number for summation.

A simulation was conducted to compare the total amplitude generated by slurry with different concentrations. In a unit period, slurry with higher sand concentration generates a higher number of impacts. Assuming sand and water were mixed uniformly in the slurry, higher sand concentration also results in a shorter interval between impacts. Compared to the simulation in Chen (2025a), the effects of both sand concentration on the total number of impacts and impact interval were considered in current study.

Five cases with different concentrations are compared using the sample signal. The total particle numbers of five cases have a ratio of 15:16:17:18:19, which is the same as the concentration ratio used in the experiments (Chen 2025a). In contrast, slurry with a lower sand concentration has a longer impact interval. The impact interval (Δt) has a ratio of 1/15:1/16:1/17:1/18:1/19 for five cases from 0.15 to 0.19 ppg of concentration. The same signal processing method used for slurry data is applied to the time domain signal. The results are then converted to a dB scale and normalized with the peak amplitude of 0.19 ppg. Figure 15 shows the comparison between cases.

The spectra in Figure 15 show the same frequency peak location at 3800 Hz with different total particle numbers (concentrations). The signal generated by the summation of more particles results in a higher total amplitude, which agrees with the observation from experimental data shown in Chen (2025a), introducing various impact intervals has no effect on the peak location.

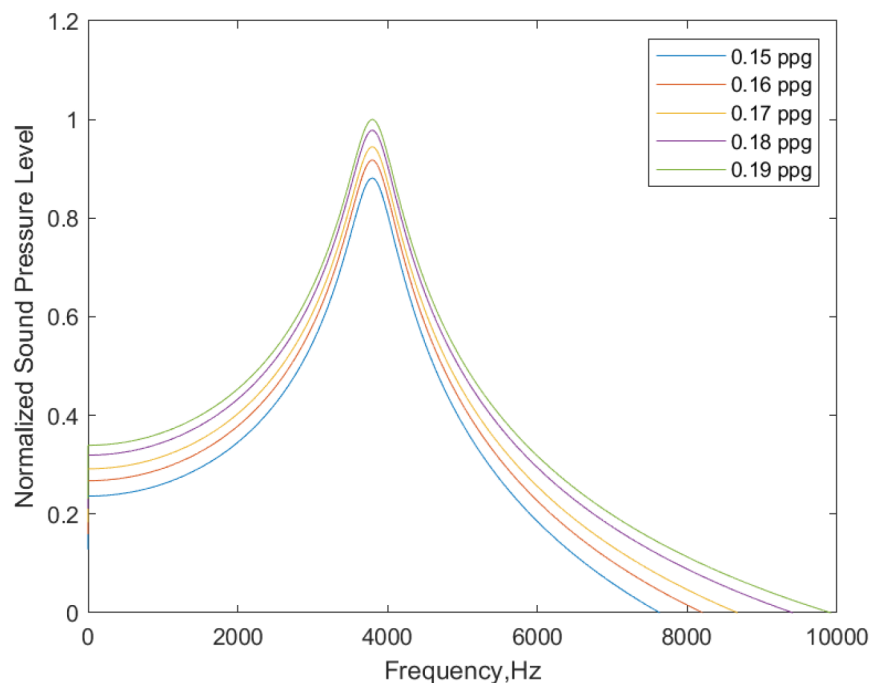


Figure 15—Comparison between Five Simulated Cases with Different Sand Concentrations

Figure 16 shows the normalized amplitude of the experimental data, the sound pressure level of all frequencies was normalized by the amplitude of the frequency peak of 0.19 ppg slurry at 3900 Hz (50.0799 dB) as shown in Figure 14. The agreement between the model and experiments indicates that the increasing spectra amplitude around 3900 Hz is caused by the sand concentration increase.

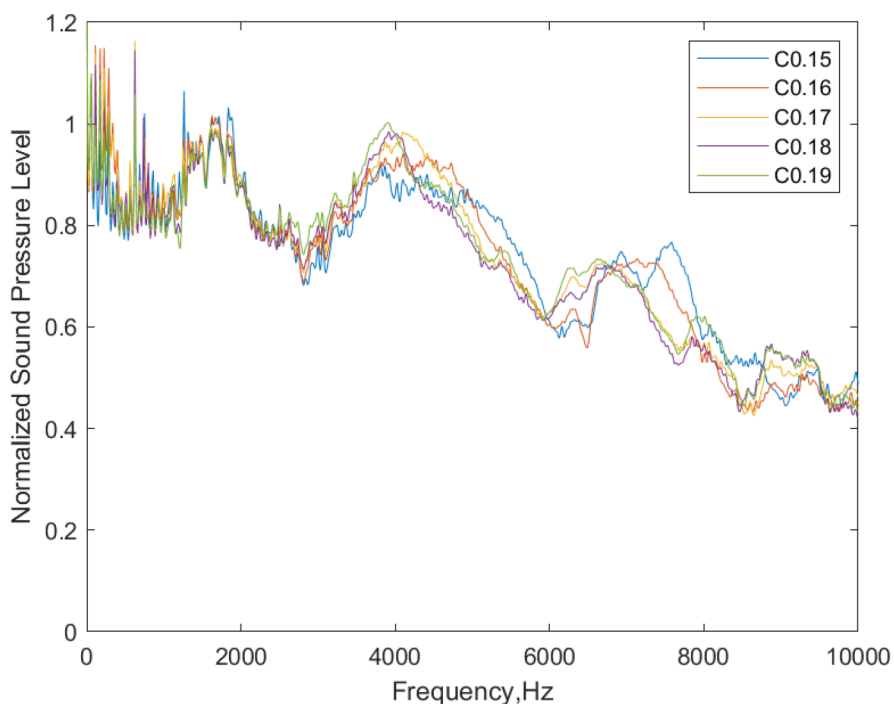


Figure 16—Normalized Spectra for Slurry with Different Sand Concentrations

When sand concentration increases, the interaction between sand particles becomes noticeable. Before the collision between sand particles and casing, the collision between sand particles can result in the dissipation

of the kinetic energy of the sand particles, which further reduces the magnitude of sound generated by sand-casing collision. Although the sound generated by two sand particle collisions is tiny, the interaction between numerous sand particles can become a new sound source other than the sounds generated by turbulent fluid and sand-casing collision. Overlapping of each sand-casing collision signal simulates an ideal case under a low sand concentration. The effect of sand particle interaction should be added to the simulation when simulating a field case in the future.

Determine the natural frequencies in the field

For sand-casing collision acoustic signal generated at the lab conditions, previous case study showed that the approach combining Hertz contact time and the system's natural frequency can predict the major frequency components effectively. This approach can also be applied to the field case. The excitation frequency range can be estimated with the same methods. However, estimating the natural frequency for the casing can be difficult. The hammer tap test cannot be directly applied to casing buried underground. The numerical frequency estimation method can still be used. The major uncertainties are casing geometry and the boundary conditions. Based on well trajectory, casing sections at different locations can have their unique deformation to fit inside the wellbore drilled.

A common method to estimate the casing geometry utilizes the torque and drag calculation when running the casing. Common inputs include the local dogleg of the well, pipe moment of inertia, Modulus, and distance between casing and wellbore. Common casing deformations include sinusoidal buckling and helical buckling.

For casing buried underground, boundary conditions are mainly determined by the cement job quality, which can be very hard to control. One potential solution to determine the boundary condition is to run a cement bond log for the entire wellbore, estimate the in-situ cement condition, and apply more realistic boundary conditions to the finite element casing model. With the casing geometry and boundary conditions determined, and the material properties of the casing can be obtained for natural frequency analysis. This approach requires casing deformation simulation to acquire geometry and a high-resolution cement bond log to acquire the detailed boundary condition. Finite element simulation on large-scale casing models can also be difficult to perform and will require strong computational power.

To obtain the natural frequency from the modal testing, without hammer impact, an alternative excitation source is required. This excitation source should meet three basic requirements:

1. The duration of excitation should be short, and the magnitude should be large enough for response capturing.
2. The excitation should be acting on the casing (near perforation) and can happen in the downhole condition.
3. Recording of the acoustic signal is available during the excitation.

With the permanent fiber installed at the sample well. High-frequency DAS signal was available during hydraulic fracturing operation. By checking the DAS signal recorded, when using a wireline perforation gun to perforate the casing, a strong acoustic signal is generated and recorded by the fiber. The location where a strong acoustic signal appears is commonly used to determine the location of the perforation. By comparing fiber response location with the wireline record, the offset between operation and designed perforation location can be determined.

Perforation guns use explosive-shaped charges to concentrate energy in a certain direction, creating high-speed high-temperature metal jets to penetrate both casing and cement. The perforating meets all the requirements of a valid excitation and can be regarded as an excitation source. By only focusing on the data at the perforation location, the time domain data includes both excitation and system response. Similar to the tap test in the lab, natural frequencies can be extracted from the FRF using frequency domain

analyses. As discussed, natural frequencies were found to be close to the sand-related frequencies since natural frequencies were triggered by the sand-casing collision during slurry injection.

Cluster selection

From the field data interpretation presented in [Chen et al., \(2025b\)](#), the spectrogram of the sample cluster showed different responses from the water injection period (T1) compared to early slurry injection (T2) and late slurry injection (T3) periods. The treatment record shows that T1 included a tiny amount of proppant, which is negligible compared to the slurry amount in T2 and T3, a full record including three periods is shown in [Figure 17](#).

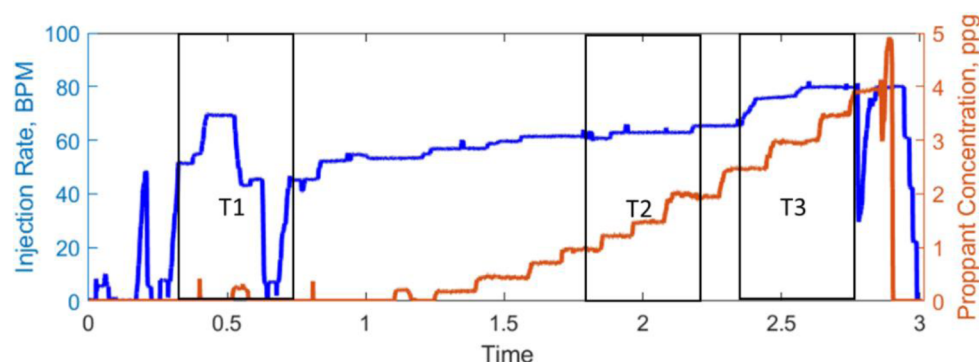


Figure 17—Three time periods during operation

A comparison of the averaged spectrum in each period is shown in [Figure 18](#).

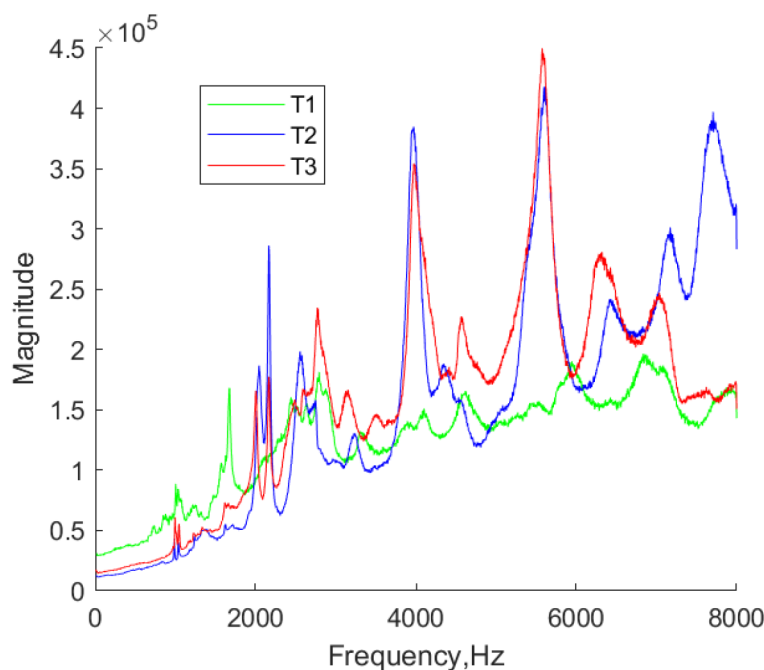


Figure 18—Averaged spectrum functions for three periods

For high-frequency DAS data recorded from the sample well, each channel in the data corresponds to a certain location. Since the sample cluster only contains 1 perforation, the channel closest to the perforation was selected.

High-frequency DAS has a sampling rate of 16000 Hz. The duration of the data is 30 seconds, which gives 480000 amplitude points in total for each channel. Figure 19 shows the time-domain acoustic data recorded at the perforation location in a full view and a zoomed-in view.

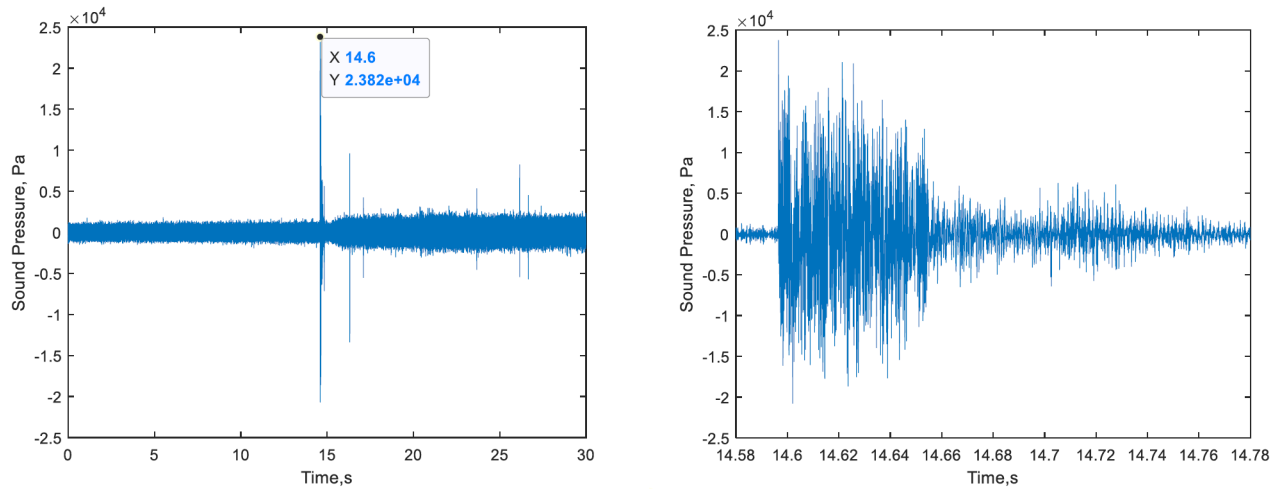


Figure 19—Amplitude Recorded at Perforation Location - Full View (Left) and a Zoom-in View (Right)

A huge peak around 14.6 seconds can be observed in Figure 19, which was generated by the high-speed metal jet. Several spikes can be observed later. The sources of these spikes are unknown. The next perforation was initiated one minute after the current perforation, which cannot be recorded in 30-second data. Short after the perforation gun fired, background noise became louder, which can be observed from 17 to 30 seconds in Figure 19. One possible reason is that the new tunnel connecting casing and formation might change the acoustic wave propagation from casing to the fiber. From the zoomed-in view of Figure 19, it can be observed that the amplitude decays with time. However, compared to the lab data shown in Figure 4, the field data did not show a clear exponential decay. The section from 14.6 to 14.78 seconds was selected for spectrum analysis.

Spectrum Analysis

Figure 20 shows the spectrum of the signal segment from 14.6 to 14.78 seconds. Several peaks can be identified, peak locations were marked on the right side.

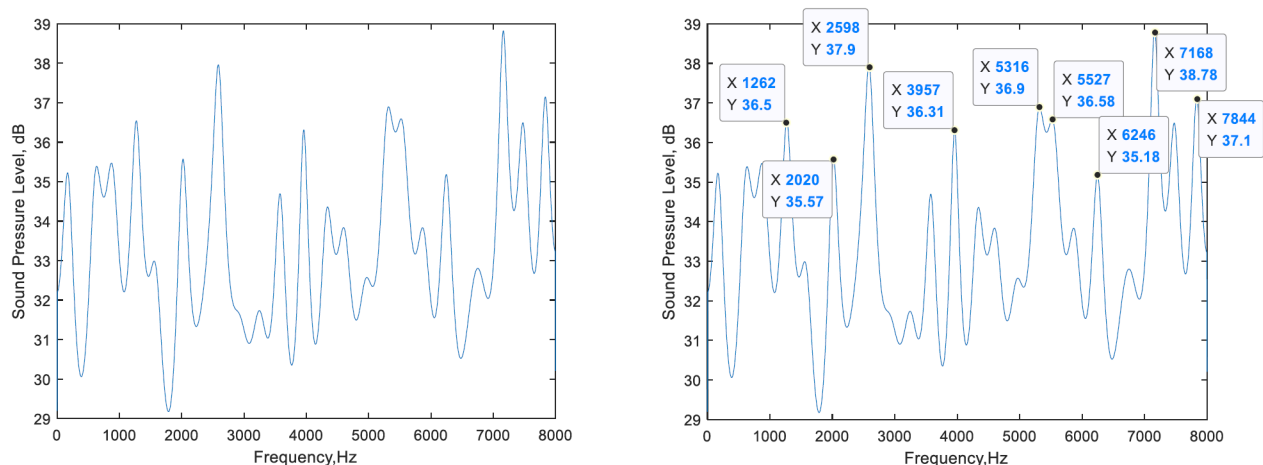


Figure 20—Spectrum of the Vibration Signal at the Perforation Location- Original Spectrum (Left) and with Peak Location Marked (Right)

Compared the casing vibration spectrum (Figure 20) to spectra of T2 and T3 periods collected during slurry injection (Figure 18), it can be observed that three spectra contained several common frequency peaks. Common peaks can be found around 2000, 3900, 5500 and 7100 Hz. From Figures 25 and 26, the top two peaks with the highest amplitude appear around 3900 and 5500 Hz. However, frequency peaks from Figure 24 do not show such a trend.

Major frequency components validation

To verify if background noise resulted in any major frequency peaks appearing in Figure 20, one period before perforating was extracted and analyzed. Figures 22 shows the signal in the time domain (left) and the spectrum (right) for the background noise extracted from 5 to 10 seconds.

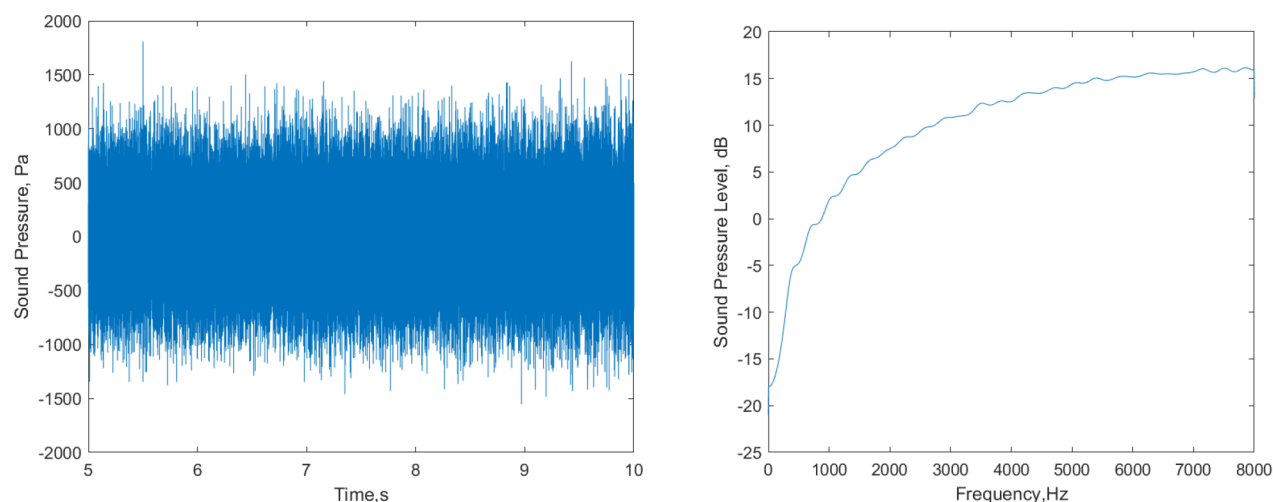


Figure 22—Background noise signal in the time domain (left) and frequency domain (right)

The signal shown in Figure 22 appears to be white noise; no clear peak can be identified from the spectrum, which indicates background noise does not contribute to the peaks found in Figure 20.

The spectrum functions in the adjacent 2 channels were also analyzed to explore the effect of location on the perforating signal captured. Figure 23 shows the signal in the time domain (left) and the spectrum (right) for the signal recorded at the adjacent channel towards the heel side. Spectrum was generated over the same time period from 14.6 to 14.78 seconds.

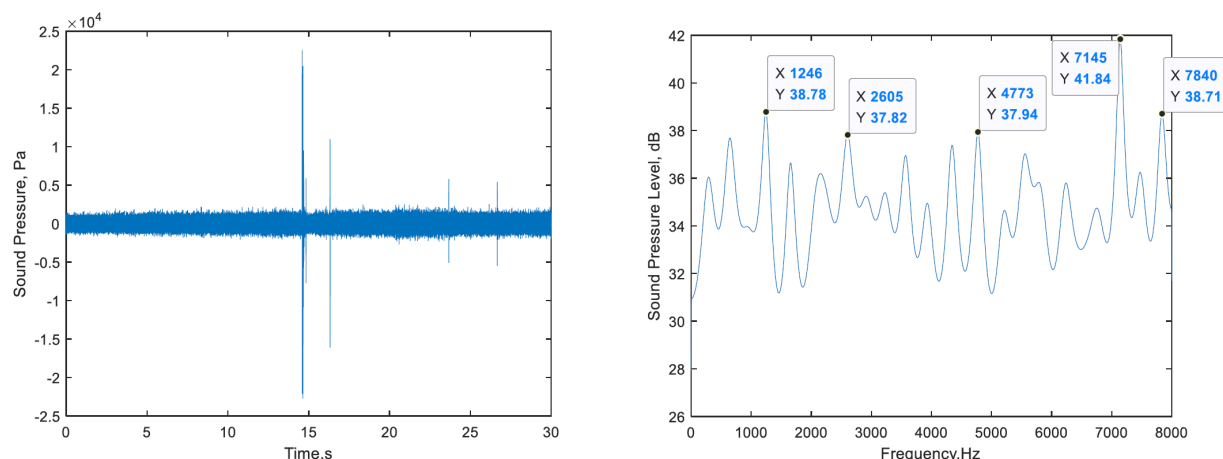


Figure 23—Acoustic Signal Recorded at the First Adjacent Channel in the Time Domain (left) and the Frequency Domain (right)

Figure 24 shows the signal in the time domain (left) and the spectrum (right) for the signal recorded at the adjacent channel towards the toe side.

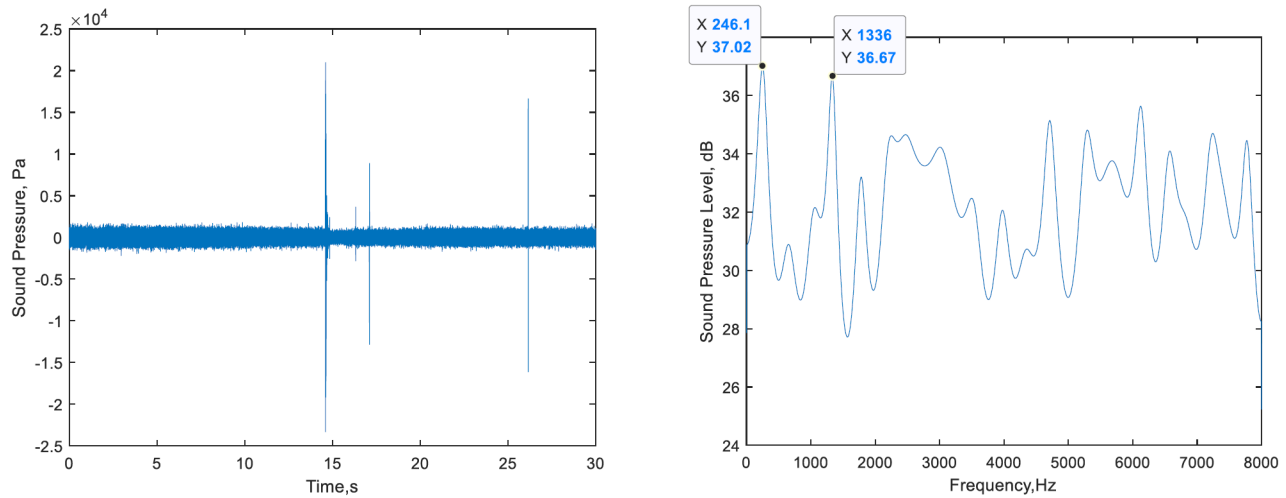


Figure 24—acoustic signal recorded at the second adjacent channel in time domain (left) and frequency domain (right)

With 1 meter away from the center channel at the perforation hole, the spectra of heel side and toe side channels (Figures 23 and 24) are different from Figure 20, very few common frequency peaks can be obtained from three figures, which indicates the spectrum is highly location related. Even with 1 meter in distance, the casing portion vibrates at different frequencies.

Conclusion

1. A theoretical approach of combining excitation frequencies with the natural frequencies of a system was developed to determine the characteristics of the acoustic signal generated from sand-casing collision and subsequent casing vibration. Hertz contact theory was applied to determine the contact time between sand particles and casing, which further provides the excitation range. Hammer impact test and modal analysis using the finite element method were selected to estimate the natural frequencies of a casing system.
2. Parameters including impact velocity, sand density and radius, Young's modulus and Poisson's Ratio of both sand and casing were determined to be related to the excitation frequency. Casing material and corresponding boundary conditions (fixture and stress) determine the natural frequency (response) of the system.
3. The developed approach was applied to the lab condition with the given geometry and material properties of the experimental apparatus and sand. Comparison between the natural frequencies estimated and the slurry signal shows good agreement on one frequency peak at around 3800 Hz, which builds a connection between sand-generated sound and the natural frequencies of a system by confirming that the impact of sand triggers the natural frequencies of the casing.
4. Based on the two excitation impulses model developed from Hertz contact theory, although the estimated excitation ranges are different, the effect of model selection on the measured response is insignificant. Current sand sizes (50 to 100 meshes) were determined to have a minor effect on the excitation spectrum. However, larger sand particles with a higher mass have more kinetic energy than small particles under the same velocity, which increases the amplitude of the acoustic signal generated by collision.
5. A new type of modal test was applied to the field data. Firing the perforating gun was regarded as the excitation source. At the perforation location, the spectrum collected after perforating and the spectra

collected during slurry injections have common frequency peak locations. Background noise does not include these frequency components and the major frequency components change significantly with location.

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Appendix A

Derivation of the impact time for an elastic collision

For two sphere particles in a collision. The contact force F has the opposite direction to the moving direction. The distance between two spheres is S , and velocities for two spheres to move forward to each other are V_1 and V_2 for sphere 1 and sphere 2. Accelerations are a_1 and a_2 for spheres 1 and 2. Masses are m_1 and m_2 for spheres 1 and 2. Then the total acceleration can be expressed as:

$$a_1 + a_2 = -\frac{F}{m_1} - \frac{F}{m_2} = -F \left(\frac{m_1 + m_2}{m_1 m_2} \right) \quad (\text{A-1})$$

By solving the deformation for a force applied on the boundary of a semi-infinite solid, the distance between two spheres under elastic and quasi-static contact can be expressed as (Timoshenko et al., 1970):

$$S = \sqrt[3]{\frac{9\pi^2 F^2 (k_1 + k_2)^2 (R_1 + R_2)}{16 R_1 R_2}} \quad (\text{A-2})$$

Effective stiffness is expressed as:

$$k_1 = \frac{1 - \nu_1^2}{\pi E_1} \text{ and } k_2 = \frac{1 - \nu_2^2}{\pi E_2} \quad (\text{A-3})$$

R_1, R_2 are the radius for spheres 1 and 2. ν_1, ν_2 are the Poisson's ratios for spheres 1 and 2. E_1, E_2 are the Young's modulus for spheres 1 and 2. Then the contact force can be expressed as:

$$F = S^{\frac{3}{2}} \sqrt{\frac{16 R_1 R_2}{9\pi^2 (k_1 + k_2)^2 (R_1 + R_2)}} \quad (\text{A-4})$$

Insert Equation A-4 to Equation A-1 gives:

$$a_1 + a_2 = -S^{\frac{3}{2}} \sqrt{\frac{16 R_1 R_2}{9\pi^2 (k_1 + k_2)^2 (R_1 + R_2)}} \left(\frac{m_1 + m_2}{m_1 m_2} \right) = a_{total} \quad (\text{A-5})$$

Let $\sqrt{\frac{16 R_1 R_2}{9\pi^2 (k_1 + k_2)^2 (R_1 + R_2)}}$ be the intermediate variable A related to stiffness and sphere radius. Let $\left(\frac{m_1 + m_2}{m_1 m_2} \right)$ be the intermediate variable B related to the mass of sphere. Mass can be calculated based on a sphere volume function as:

$$m_1 = \frac{4}{3} \pi R_1^3 \rho_1 \text{ and } m_2 = \frac{4}{3} \pi R_2^3 \rho_2 \quad (\text{A-6})$$

Where ρ_1, ρ_2 are the densities for spheres 1 and 2. With intermediate A and B , total acceleration from Equation A-5 can be expressed as:

$$a_{total} = -S^{\frac{3}{2}} AB \quad (\text{A-7})$$

Multiply both sides by $V_{total} = V_1 + V_2 = \frac{dS}{dt}$, and $a_{total} = \frac{dV_{total}}{dt}$, then do integration on both sides, which gives:

$$\begin{aligned} \int \frac{dV_{total}}{dt} V_{total} &= - \int ABS^{\frac{3}{2}} \frac{dS}{dt} \\ \int V_{total} dV_{total} &= - \int ABS^{\frac{3}{2}} dS \\ \frac{1}{2} V_{total}^2 + C &= -AB \frac{2}{5} S^{\frac{5}{2}} \end{aligned} \quad (\text{A-8})$$

Applying the first boundary condition that the distance between two spheres decreases to zero when a collision just happens, which is $\frac{1}{2}V_{total}^2 + C = 0$, therefore $C = -\frac{1}{2}V_{impact}^2$. Applying the second boundary condition: when the total velocity decreases from the impact velocity to zero, the kinetic energy transfers to strain energy. At this moment, two spheres experience the maximum deformation and contact pressure. Maximum deformation appears when V_{total} decreases to zero: $-\frac{1}{2}V_{impact}^2 = -AB\frac{2}{5}S^{\frac{5}{2}}$ and $S_{maximum} = \left(\frac{5}{4}\frac{V_{impact}^2}{AB}\right)^{\frac{2}{5}}$, by plugging the first and the second boundary conditions into Equation A-8:

$$\frac{1}{2}V_{total}^2 = \frac{1}{2}V_{impact}^2 - AB\frac{2}{5}S^{\frac{5}{2}} \quad (A-9)$$

$$\begin{aligned} \frac{dS}{dt} &= V_{total} = \sqrt{V_{impact}^2 - \frac{4}{5}AB\frac{2}{5}S^{\frac{5}{2}}} \\ dt &= \frac{1}{\sqrt{V_{impact}^2 - \frac{4}{5}AB\frac{2}{5}S^{\frac{5}{2}}}} dS = \frac{1}{V_{impact}} * \frac{1}{\sqrt{1 - \frac{1}{V_{impact}^2} \frac{4}{5}AB\frac{2}{5}S^{\frac{5}{2}}}} dS \end{aligned} \quad (A-10)$$

Let $\frac{4}{5}\frac{AB}{V_{impact}^2}S^{\frac{5}{2}} = D^{\frac{5}{2}}$ to eliminate the constant term, where D is an intermediate variable.

$\left(\frac{4}{5}\frac{AB}{V_{impact}^2}\right)^{\frac{2}{5}} = \frac{D}{S} = \frac{1}{S_{maximum}}$. Therefore, D can also be regarded as the ratio between the current deformation and the maximum deformation. $dD = \left(\frac{5}{4}\frac{V_{impact}^2}{AB}\right)^{\frac{2}{5}} dS$. The original function integrates from no deformation ($S = 0$) to maximum deformation ($S = S_{maximum}$). With ratio D , the new function integrates from $D=0$ to $D=1$, which gives:

$$dt = \left(\frac{5}{4}\frac{V_{impact}^2}{AB}\right)^{\frac{2}{5}} * \frac{1}{V_{impact}} * \int_0^1 \frac{1}{\sqrt{1 - D^{\frac{5}{2}}}} dD \quad (A-11)$$

$$t = \left(\frac{5}{4}\frac{V_{impact}^2}{AB}\right)^{\frac{2}{5}} * \frac{1}{V_{impact}} * 1.47 \quad (A-12)$$

Time t represents the period from when the collision happens to the deformation reaching the maximum level. Since there is no energy lost in kinetic energy-strain energy transformation, and vice versa, elastic collision assumes it takes the same time for maximum deformation to decrease back to zero where two spheres are no longer in contact. The total impact time can be expressed as:

$$t_{el} = \left(\frac{5}{4}\frac{V_{impact}^2}{AB}\right)^{\frac{2}{5}} * \frac{1}{V_{impact}} * 1.47 * 2 \quad (A-13)$$

For a two-sphere system, the intermediate variable A is set as:

$$A = \sqrt{\frac{16}{9\pi^2} \frac{R_1 R_2}{(k_1 + k_2)^2 (R_1 + R_2)}} \quad (A-14)$$

In the case that a collision happens between a plate and a sphere particle, a plate can be regarded as a sphere with an infinite radius. Intermediate variable A for a plate can be expressed as:

$$A_{plate} = \lim_{R_1 \rightarrow \infty} \sqrt{\frac{16}{9\pi^2} \frac{R_1 R_2}{(k_1 + k_2)^2 (R_1 + R_2)}} = \sqrt{\frac{16R_2}{9\pi^2 (k_1 + k_2)^2}} \quad (A-15)$$

Intermediate variable B for a two-sphere collision system is set as:

$$B = \left(\frac{m_1 + m_2}{m_1 m_2} \right) = \frac{\frac{4}{3} (\pi R_1^3 \rho_1 + \pi R_2^3 \rho_2)}{\left(\frac{4}{3} \right)^2 \pi^2 R_1^3 \rho_1 R_2^3 \rho_2} = \frac{R_1^3 \rho_1 + R_2^3 \rho_2}{\frac{4}{3} \pi R_1^3 \rho_1 R_2^3 \rho_2} \quad (A-16)$$

Intermediate variable B for a plate can be expressed as:

$$B_{plate} = \lim_{R_1 \rightarrow \infty} \frac{R_1^3 \rho_1 + R_2^3 \rho_2}{\frac{4}{3} \pi R_1^3 \rho_1 R_2^3 \rho_2} = \frac{R_1^3 \rho_1}{\frac{4}{3} \pi R_1^3 \rho_1 R_2^3 \rho_2} = \frac{3}{4\pi R_2^3 \rho_2} \quad (A-17)$$

With A_{plate} and B_{plate} , the product is:

$$A_{plate} B_{plate} = \frac{4}{3\pi} \frac{R_2^{0.5}}{k_1 + k_2} * \frac{3}{4\pi R_2^3 \rho_2} = \frac{1}{(k_1 + k_2) \pi^2 R_2^{\frac{5}{2}} \rho_2} \quad (A-18)$$

The total time for a plate-sphere system is then:

$$\begin{aligned} t_{el} &= \left(\frac{5}{4} V_{impact}^2 (k_1 + k_2) \pi^2 R_2^{\frac{5}{2}} \rho_2 \right)^{\frac{2}{5}} * \frac{1}{V_{impact}} * 1.47 * 2 \\ &= 2.94 * \left[\left(\frac{1-v_1^2}{\pi E_1} + \frac{1-v_2^2}{\pi E_2} \right) \pi^2 \rho_2 \right]^{\frac{2}{5}} R_2 V_{impact}^{\frac{4}{5}} V_{impact}^{-1} \end{aligned} \quad (A-19)$$

This can be rearranged as:

$$\begin{aligned} t_{el} &= 2.94 \left(\frac{5\pi}{4} \right)^{\frac{2}{5}} \rho_2^{\frac{2}{5}} \frac{R_2}{V_{impact}^{\frac{1}{5}}} (f(E))^{\frac{2}{5}} \\ f(E) &= \frac{1-v_1^2}{E_1} + \frac{1-v_2^2}{E_2} \end{aligned} \quad (A-20)$$

Appendix B

Force impulse and corresponding excitation spectrum

Assuming the force impulse is a triangular impulse, the corresponding spectrum can be obtained with the following steps:

For a square impulse, an amplitude A and duration of τ can be expressed as:

$$f(t) = A \quad \text{on} \quad \left[-\frac{\tau}{2}, \frac{\tau}{2}\right] \quad (\text{B-1})$$

Fourier transforms of the square impulse can be expressed as:

$$F(\omega) = \int_{-\infty}^{+\infty} f(t)e^{-j\omega t} dt = \int_{-\frac{\tau}{2}}^{+\frac{\tau}{2}} Ae^{-j\omega t} dt \quad (\text{B-2})$$

With the Euler equation:

$$\begin{aligned} F(\omega) &= \int_{-\frac{\tau}{2}}^{+\frac{\tau}{2}} A(\cos\omega t - j\sin\omega t) dt \\ &= \int_{-\frac{\tau}{2}}^{+\frac{\tau}{2}} \cos\omega t dt - A \int_{-\frac{\tau}{2}}^{+\frac{\tau}{2}} j\sin\omega t dt \end{aligned} \quad (\text{B-3})$$

$Aj\sin\omega t$ is an odd function, integration of an odd function over a period $\left(-\frac{\tau}{2}, \frac{\tau}{2}\right)$ equals 0, which only leaves:

$$\begin{aligned} F(\omega) &= A \int_{-\frac{\tau}{2}}^{+\frac{\tau}{2}} \cos\omega t dt = 2A \int_0^{+\frac{\tau}{2}} \cos\omega t dt \\ &= \frac{2A}{\omega} \sin\omega t \Big|_0^{+\frac{\tau}{2}} = \frac{2A}{\omega} \sin \frac{\omega\tau}{2} \end{aligned} \quad (\text{B-4})$$

Which can be written as a sinc function where $\text{sinc } x = \frac{\sin x}{x}$

$$F(\omega) = A\tau \frac{\sin \frac{\omega\tau}{2}}{\frac{\omega\tau}{2}} = A\tau \text{sinc} \frac{\omega\tau}{2} \quad (\text{B-5})$$

The spectrum function of the triangular impulse generated by taking the convolution of the square impulse is then $F(\omega)_T = F(\omega)_S^2 = A^2\tau^2 \text{sinc}^2 \frac{\omega\tau}{2}$. $F(\omega)_T$ and $F(\omega)_S$ are spectrum functions of triangular and square impulses. The convolution of two square impulses with the same duration of τ gives a triangular impulse with a duration of 2τ , and the area under the triangular impulse equals the product of the area of two square impulses, the height of the triangular impulse is:

$$(A\tau)^2 \div 2\tau = \frac{A^2\tau}{2} \quad (\text{B-6})$$

Let the new duration of triangular impulse be $\tau' = 2\tau$, the new amplitude be $A' = \frac{A^2\tau}{2} = \frac{A^2\tau'}{4}$. For a triangular impulse with a duration of τ' and amplitude of A' , the Fourier transform equals

$$A^2\tau^2 \text{sinc}^2 \frac{\omega\tau}{2} = A'\tau' \text{sinc}^2 \frac{\omega\tau'}{4} = \frac{A'\tau'}{2} \text{sinc}^2 \frac{\omega\tau'}{4} \quad \text{or} \quad \frac{A'\tau'}{2} \left(\frac{\sin \frac{\omega\tau'}{4}}{\frac{\omega\tau'}{4}} \right)^2.$$

Assuming the force impulse is a half-sine impulse, the corresponding spectrum can be obtained with the following steps:

With the Euler equation:

$$\begin{aligned}
 F(\omega) &= A \int_{-\tau/2}^{\tau/2} \cos \frac{\pi}{\tau} t e^{-j\omega t} dt \\
 &= \frac{A}{2} \int_{-\tau/2}^{\tau/2} (e^{j\frac{\pi}{\tau} t} + e^{-j\frac{\pi}{\tau} t}) e^{-j\omega t} dt \\
 &= \frac{A}{2} \left[\int_{-\tau/2}^{\tau/2} e^{-j(\omega - \frac{\pi}{\tau}) t} dt + \int_{-\tau/2}^{\tau/2} e^{-j(\omega + \frac{\pi}{\tau}) t} dt \right] \\
 &= \frac{A}{2} \left[\frac{1}{-j(\omega - \frac{\pi}{\tau})} e^{-j(\omega - \frac{\pi}{\tau}) t} \Big|_{-\tau/2}^{\tau/2} + \frac{1}{-j(\omega + \frac{\pi}{\tau})} e^{-j(\omega + \frac{\pi}{\tau}) t} \Big|_{-\tau/2}^{\tau/2} \right]
 \end{aligned} \tag{B-7}$$

With the reverse Euler equation:

$$\begin{aligned}
 F(\omega) &= \frac{A}{2} \left[\frac{e^{-j(\omega - \frac{\pi}{\tau}) \frac{\tau}{2}} - e^{-j(\omega - \frac{\pi}{\tau}) (-\frac{\tau}{2})}}{-j(\omega - \frac{\pi}{\tau})} + \frac{e^{-j(\omega + \frac{\pi}{\tau}) \frac{\tau}{2}} - e^{-j(\omega + \frac{\pi}{\tau}) (-\frac{\tau}{2})}}{-j(\omega + \frac{\pi}{\tau})} \right] \\
 &= A \left[\frac{\sin(\frac{\omega\tau - \pi}{2})}{\omega - \frac{\pi}{\tau}} + \frac{\sin(\frac{\omega\tau + \pi}{2})}{\omega + \frac{\pi}{\tau}} \right]
 \end{aligned} \tag{B-8}$$

The same result can also be obtained from the Fourier transform of multiplication of a cosine function and a square impulse with height of 1 from $-1/2\tau$ to $1/2\tau$. The Fourier transform of cosine is:

$$F[A \cos \frac{\pi}{\tau} t] = A\pi \delta(\omega - \frac{\pi}{\tau}) + A\pi \delta(\omega + \frac{\pi}{\tau}) \tag{B-9}$$

As discussed previously, the Fourier transform of a square impulse is $\tau \frac{\sin \frac{\omega\tau}{2}}{\frac{\omega\tau}{2}}$.

$$\begin{aligned}
 F[x_1(t) \cdot x_2(t)] &= \frac{1}{2\pi} X_1(\omega) * X_2(\omega) \\
 &= \frac{1}{2\pi} [A\pi \delta(\omega - \frac{\pi}{\tau}) + A\pi \delta(\omega + \frac{\pi}{\tau})] * \tau \frac{\sin \frac{\omega\tau}{2}}{\frac{\omega\tau}{2}}
 \end{aligned} \tag{B-10}$$

Convolution resulted in frequency shifts of π/τ in negative and positive directions:

$$\begin{aligned}
 \frac{1}{2\pi} X_1(\omega) * X_2(\omega) &= A\pi \frac{1}{2\pi} \tau \left[\frac{\sin(\frac{(\omega - \frac{\pi}{\tau})\tau}{2})}{\frac{(\omega - \frac{\pi}{\tau})\tau}{2}} + \frac{\sin(\frac{(\omega + \frac{\pi}{\tau})\tau}{2})}{\frac{(\omega + \frac{\pi}{\tau})\tau}{2}} \right] \\
 &= A \left[\frac{\sin(\frac{\omega\tau - \pi}{2})}{\omega - \frac{\pi}{\tau}} + \frac{\sin(\frac{\omega\tau + \pi}{2})}{\omega + \frac{\pi}{\tau}} \right]
 \end{aligned} \tag{B-11}$$